

PORTFOLIO PROJECT • DATA ANALYTICS

Customer Shopping Behaviour Analysis

Turning 3,900 customer records into marketing and product decisions

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Python • SQL (PostgreSQL) • Power BI



Python

Data preparation



PostgreSQL

Business questions



Power BI

Interactive dashboard

What the business wants to know



How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?

Business question from the project brief



Who drives revenue?

How spending differs by gender and age group



What sells best?

Top categories, products and customer ratings



What influences spending?

The role of discounts, shipping type and reviews



What builds loyalty?

Repeat buyers, segments and subscriptions

3,900 customers, analysed end to end

3,900

customers, one row each

18

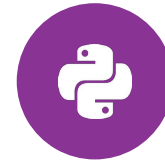
columns: demographics,
purchase details, loyalty

25

products across 4 categories

37

missing values (Review
Rating only)



Prepare Python (pandas)

Clean, fix missing values, add features



Analyse SQL (PostgreSQL)

Answer 10 business questions



Visualise Power BI

Interactive dashboard with slicers



Communicate Report and deck

Findings and recommendations

Preparing the data for analysis



01

Fill missing ratings

37 gaps filled with the median rating of the same category



02

Standardise names

Columns renamed to snake_case for easy SQL



03

Add age groups

Young Adult, Adult, Middle-aged, Senior (quartiles)



04

Add purchase frequency

Labels like 'Monthly' converted to days (30)



05

Remove duplicate column

promo_code_used matched discount_applied in every row



06

Load to PostgreSQL

Clean table written to the customer table

Result: 3,900 rows • 19 columns • 0 missing values, ready for SQL and Power BI

KEY INSIGHT

Revenue follows customer volume, not spend per order

\$233,081

total revenue from 3,900 customers

≈ \$60

average purchase in every customer group, from \$57 to \$61

Men generate 68% of revenue simply because they are 68% of customers. The same pattern holds for categories and age groups.

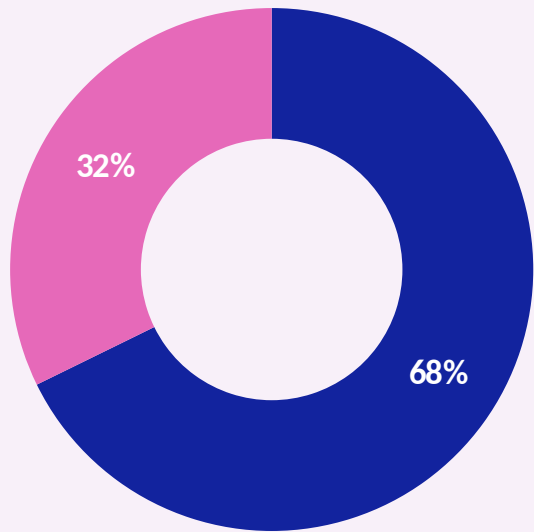
Average purchase amount by customer group (\$)



So what: growth has to come from more customers, more frequent orders or bigger baskets, not from any one segment.

Who drives revenue?

Revenue by gender



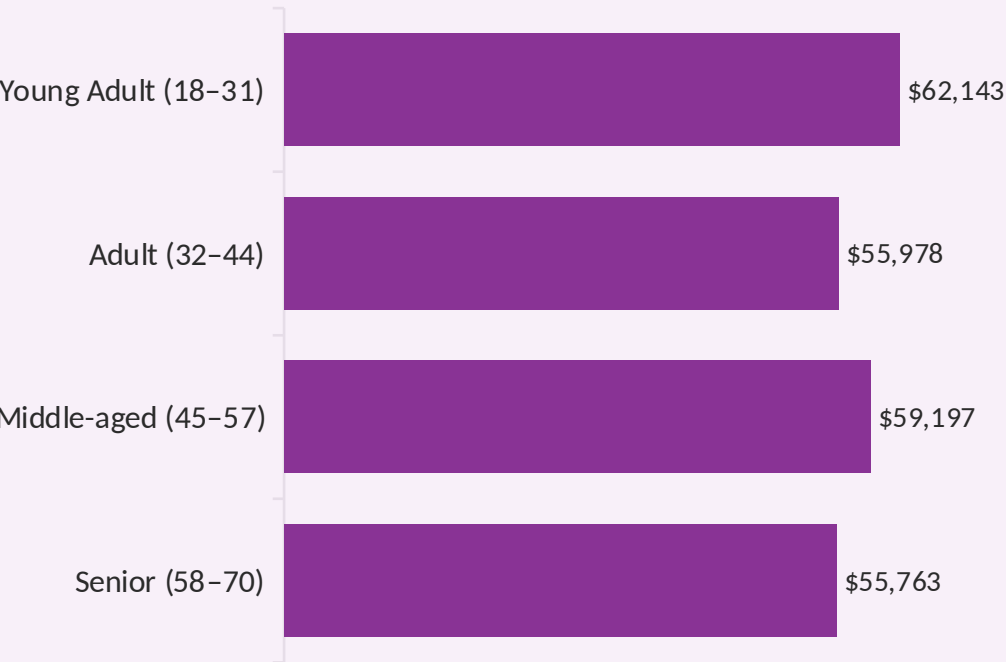
\$157,890

Male • 2,652 customers • \$59.54 per order

\$75,191

Female • 1,248 customers • \$60.25 per order

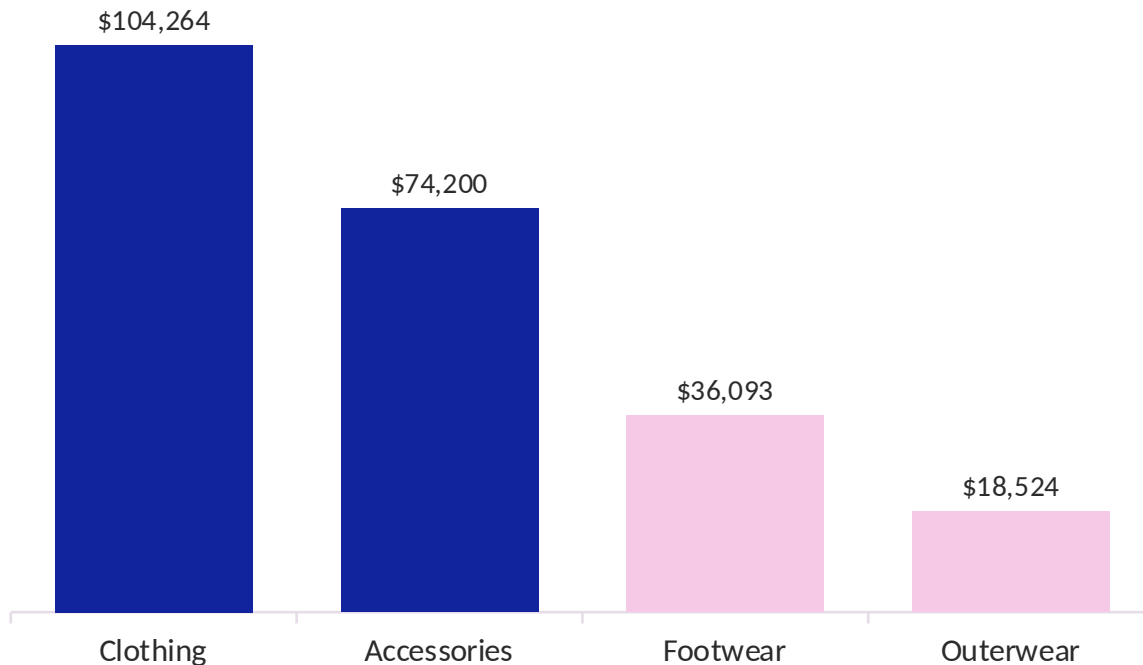
Revenue by age group



So what: men bring in two-thirds of revenue, and Young Adults lead by age, but spend per order is the same for everyone.

Clothing and Accessories carry the business

Revenue by category



Top products by orders

Clothing

44.7% of revenue

Blouse 171 · Pants 171 · Shirt 169

Accessories

31.8% of revenue

Jewelry 171 · Belt 161 · Sunglasses 161

Footwear

15.5% of revenue

Sandals 160 · Shoes 150 · Sneakers 145

Outerwear

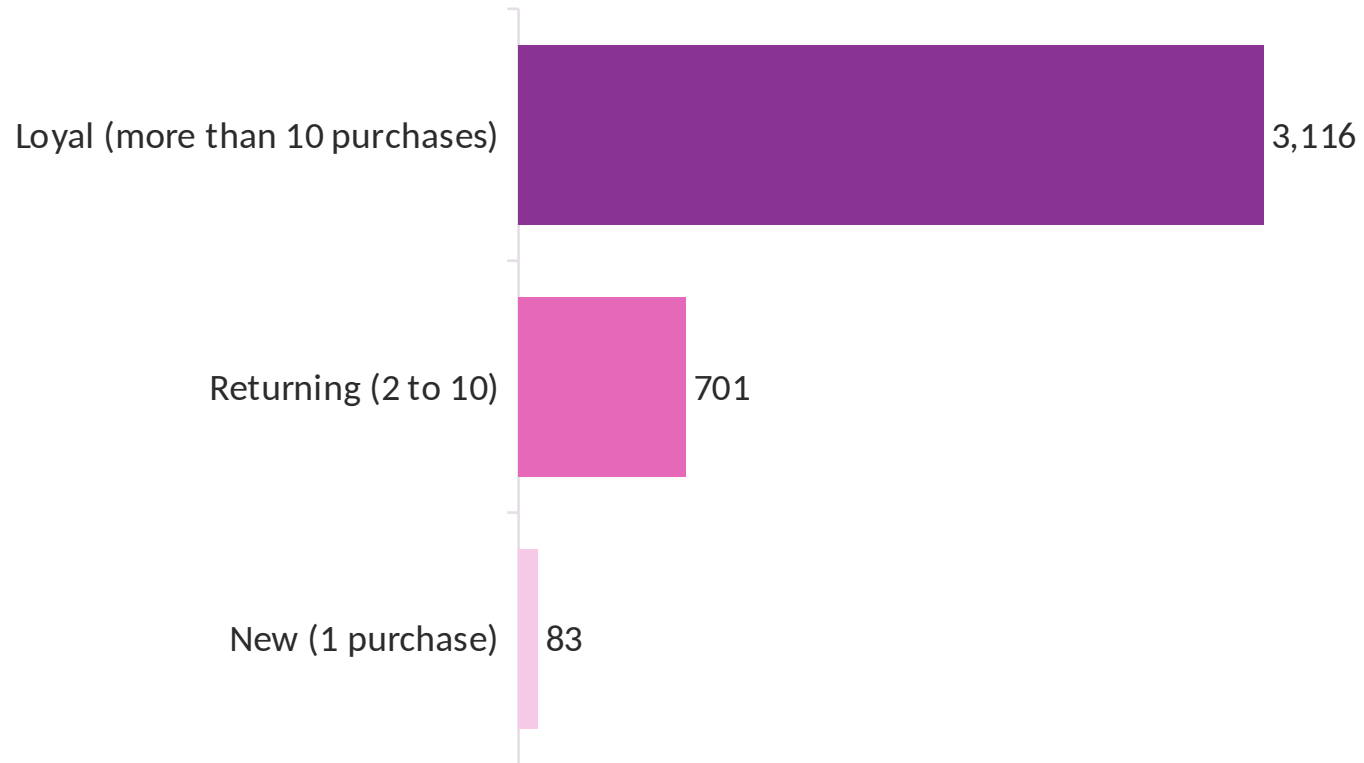
7.9% of revenue

Jacket 163 · Coat 161



So what: 76.5% of revenue comes from Clothing and Accessories; bundle them with Footwear and Outerwear to grow both.

Most customers are already loyal



80%

of customers are Loyal

25

previous purchases on average

2%

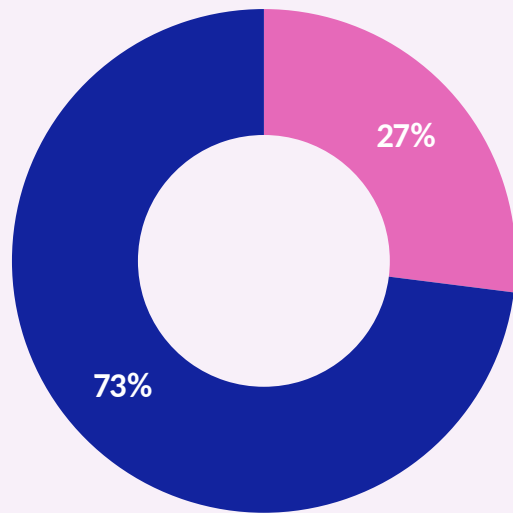
are New customers (83 people)



So what: the customer base is mature, so keeping loyal customers and raising their value matters more than acquisition alone.

The subscription is not lifting spend

Customers by subscription



■ Subscribed 1,053 ■ Not 2,847

\$59.49 vs \$59.87

average spend per order: subscribers spend slightly less

26.9% of revenue

comes from subscribers, matching their 27% share of customers

27.6% vs 22.4%

of repeat buyers subscribe, compared with other customers

2,518 repeat buyers

have not subscribed yet, the biggest opportunity



So what: rework subscription benefits to encourage bigger or more frequent orders, and target unsubscribed repeat buyers.

Discounts are common, but not only for low spenders

43%

of all purchases used a discount (1,677 customers)

839

discount users still spent above the \$59.76 average

50%

of discount users are above-average spenders

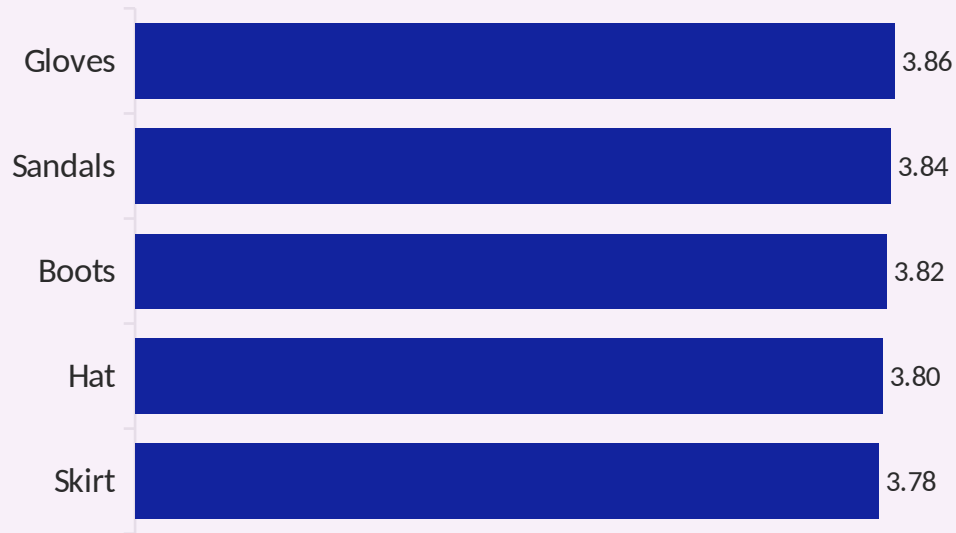
Most discount-dependent products (% of purchases discounted)



So what: review margins on the most discounted products and offer premium deals to high-spending discount users.

Ratings are average; faster shipping means slightly higher spend

Top 5 products by average rating (out of 5)



Every product averages between 3.62 (Shirt) and 3.86. Lowest rated: Shirt 3.62, Jeans 3.65, Blouse 3.68.

Average purchase by shipping type



\$60.48

Express



\$58.46

Standard

+3.5%

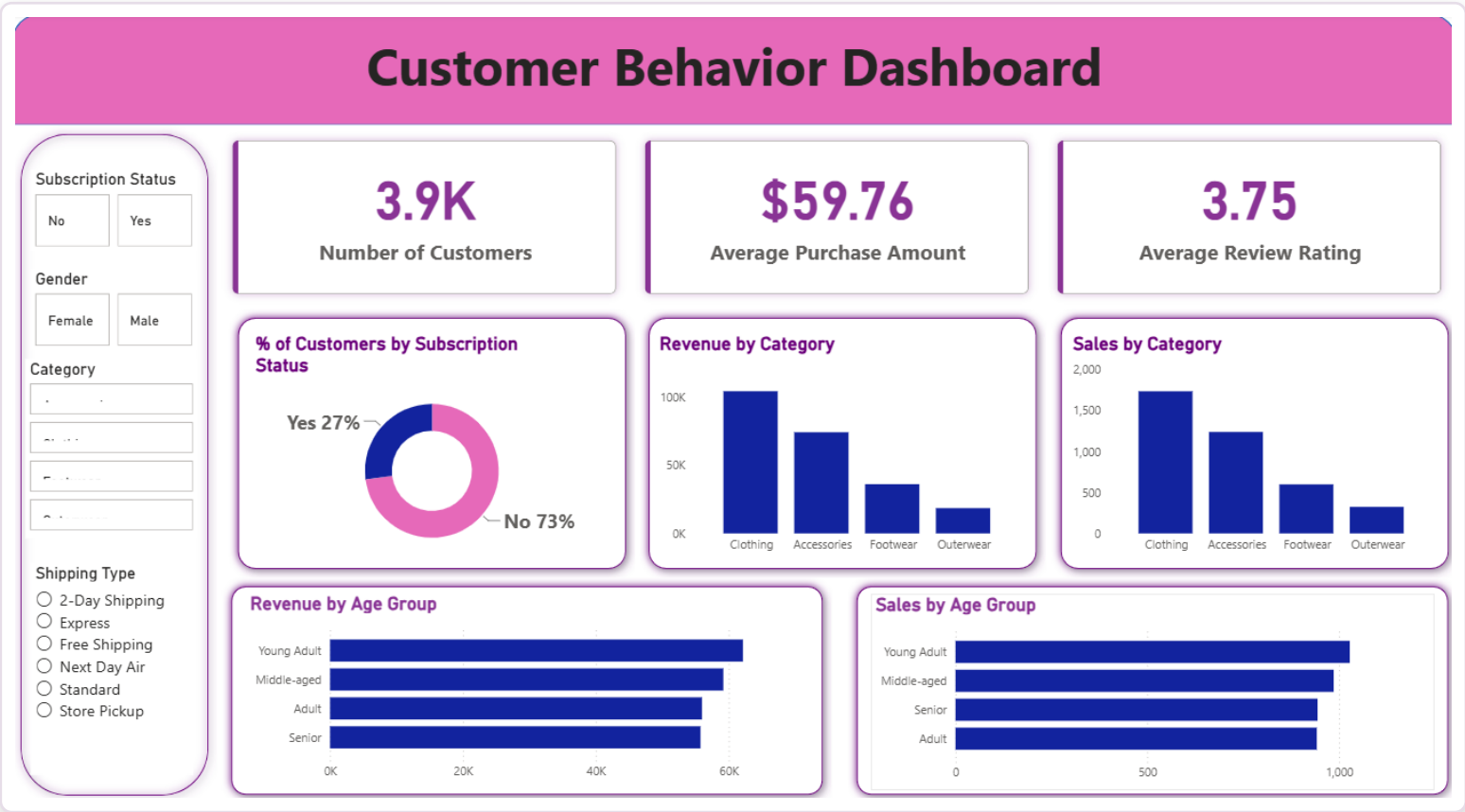
more per order with
Express

All six shipping types average between \$58.46 and \$60.73, so the difference is small.



So what: no product clearly delights customers. Use reviews to improve low-rated items, and test free express shipping as a perk.

An interactive dashboard for day-to-day decisions



3 KPI cards

Customers (3.9K), average purchase (\$59.76) and average rating (3.75)



5 charts

Subscription split, plus revenue and orders by category and by age group



4 slicers

Filter by subscription status, gender, category and shipping type



3 DAX measures

COUNT and AVERAGE measures behind the KPI cards

Connected live to the PostgreSQL customer table

Seven recommendations



Rework subscription benefits

Add perks that drive bigger or more frequent orders; target the 2,518 unsubscribed repeat buyers



Strengthen core categories

Keep Clothing and Accessories prominent; bundle Footwear and Outerwear with them



Reward loyal customers

Tiered loyalty programme for the 3,116 Loyal customers



Improve product satisfaction

Act on feedback for Shirt, Jeans, Blouse; feature Gloves, Sandals, Boots



Review the discount strategy

Check margins on Hat, Sneakers, Coat, Sweater, Pants; premium offers for 839 high spenders



Target campaigns by age

Tailor campaigns for Young Adults while keeping messaging broad



Grow average order value

Bundles, cross-selling and free-shipping thresholds to increase basket size

Start with the first three: they target the largest groups of customers and revenue.

Limitations and next steps



Limitations

Snapshot data

One latest purchase per customer and no dates, so no trends over time

Dataset quirk

No female customer subscribed or used a discount, likely a sample artefact

Even distributions

Season, payment method and size are spread almost evenly

Correlation only

Differences such as higher spend with Express do not prove cause



Next steps

Add purchase dates

Track trends and measure real repeat-purchase timing

A/B test discounts

Find the best balance between sales and margin

Predict subscriptions

Model which repeat buyers are most likely to subscribe

RFM segmentation

Use recency, frequency and monetary value once history exists

THE BOTTOM LINE

Grow revenue with more loyal customers and bigger baskets, not deeper discounts.

80%

customers already loyal

≈ \$60

spend per order everywhere

76.5%

revenue from 2 categories

Thank you



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